Nimmanagoti Manoj

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.79
Sri chaitanya junior kalasala	Class XII (Telangana state board of intermediate education)	2019	95.40%
Jaya high school	Class X (Board of secondary education,Telangana state)	2017	9.70%

PROJECTS

Human Activity Recognition:

(https://github.com/Manoj-nimmanagoti/human-activity-recognition)

- Developed a machine learning-based system to classify human activities using sensor data.
- Performed data preprocessing, feature preparation and class-imbalance handling to improve model training.
- Evaluated LSTM and Transformer architectures and compared their classification performance on unseen activity data.

Stock Price Prediction:

(https://github.com/Manoj-nimmanagoti/stock-price-prediction):

- Developed a time-series forecasting system using historical stock-market data, analyzing Open, High, Low, Close and Volume attributes.
- Implemented and compared Moving Average, Linear Regression, KNN, Prophet and LSTM approaches for stock-price prediction.
- Evaluated forecasting performance using RMSE and analyzed predicted versus actual prices.

GAN-Based Synthetic Data Generation:

(https://github.com/Manoj-nimmanagoti/synthetic-data-generation)

- Developed a Generative Adversarial Network to generate synthetic data resembling the distribution of real training samples.
- Designed and trained generator and discriminator networks using adversarial learning.
- Evaluated the generated samples against the original data to assess the quality of synthetic data.

Digital Modulation Scheme Classification using Deep Learning:

(https://github.com/Manoj-nimmanagoti/AWGN_AMC_DL)

- Developed a deep learning-based system for automatic classification of digital modulation schemes from IQ signal samples
- Generated and processed signal datasets using TorchSig and performed preprocessing of complex-valued IQ samples for model training and evaluation.
- Trained a neural network to distinguish modulation schemes including BPSK, QPSK, 8-PSK, 16-QAM and 64-QAM, and evaluated classification performance on unseen test samples.

SKILLS

Programming Languages:

C, C++, Python, Verilog

Technical Skills:

Data Structures & Algorithms, Machine Learning, Deep Learning, Natural Language Processing, SQL

Tools & Software:

MATLAB, Cadence, Quartus, LTspice, AutoCAD, Eagle

RELEVANT COURSES

Random Variables and Stochastic Processes, Matrix Theory, Principles of Digital Communication, Digital Signal Processing, Information Theory and Coding, Pattern Recognition and Machine Learning, Wireless Communication, Advanced Communication Engineering

EXTRACURRICULAR

Active participant in cricket and badminton; regularly engage in recreational and competitive matches.